

Module 04 Lab - Exploratory Data Analysis

Objective: To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model.**This lab is fully coded.** Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Part 1: Setup and Data Loading

What is Exploratory Data Analysis (EDA)? EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data. What stories does it tell? Are there errors or missing values? Are there strong relationships between variables? EDA helps answer these questions. We will use the famous Titanic dataset for this lab. It contains information about passengers and, crucially, whether they survived the disaster.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset directly from a URL
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')

print("---- First 5 Rows ----")
print(df.head())

print("---- Basic Info ----")
df.info()
```

```
---- First 5 Rows ----
   PassengerId  Survived  Pclass \
0             1         0       3
1             2         1       1
2             3         1       3
3             4         1       1
4             5         0       3

                                Name    Sex  Age  SibSp  \
0                                Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                                Heikkinen, Miss. Laina  female  26.0      0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)    female  35.0      1
4                                Allen, Mr. William Henry    male  35.0      0

   Parch    Ticket   Fare Cabin Embarked
0      0   A/5 21171   7.2500   NaN        S
1      0    PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0   113803   53.1000  C123        S
4      0   373450   8.0500   NaN        S

---- Basic Info ----
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null   int64
1   Survived        891 non-null   int64
2   Pclass          891 non-null   int64
3   Name            891 non-null   object
4   Sex             891 non-null   object
5   Age            714 non-null   float64
6   SibSp          891 non-null   int64
7   Parch          891 non-null   int64
8   Ticket         891 non-null   object
9   Fare           891 non-null   float64
10  Cabin          204 non-null   object
11  Embarked       889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Part 2: Descriptive StatisticsLet's start by getting a high-level numerical summary of the data.

- ✓ The `.describe()` method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```
# Get summary statistics for numerical columnsprint("---- Descriptive Statistics ----")print(df.describe())print("---- Key Insights ----")
```

Part 3: Visual EDA - Telling Stories with PlotsNumbers are great, but plots make patterns and relationships immediately obvious. The goal of visual EDA is to turn data into insights.**A Note on**

- ✓ **Plotting Libraries:**
 - * **Matplotlib:** The foundational library, gives you full control over everything.
 - * **Seaborn:** Built on Matplotlib, it provides a simpler, high-level interface for creating common statistical plots. We will use Seaborn for its ease of use and attractive defaults.

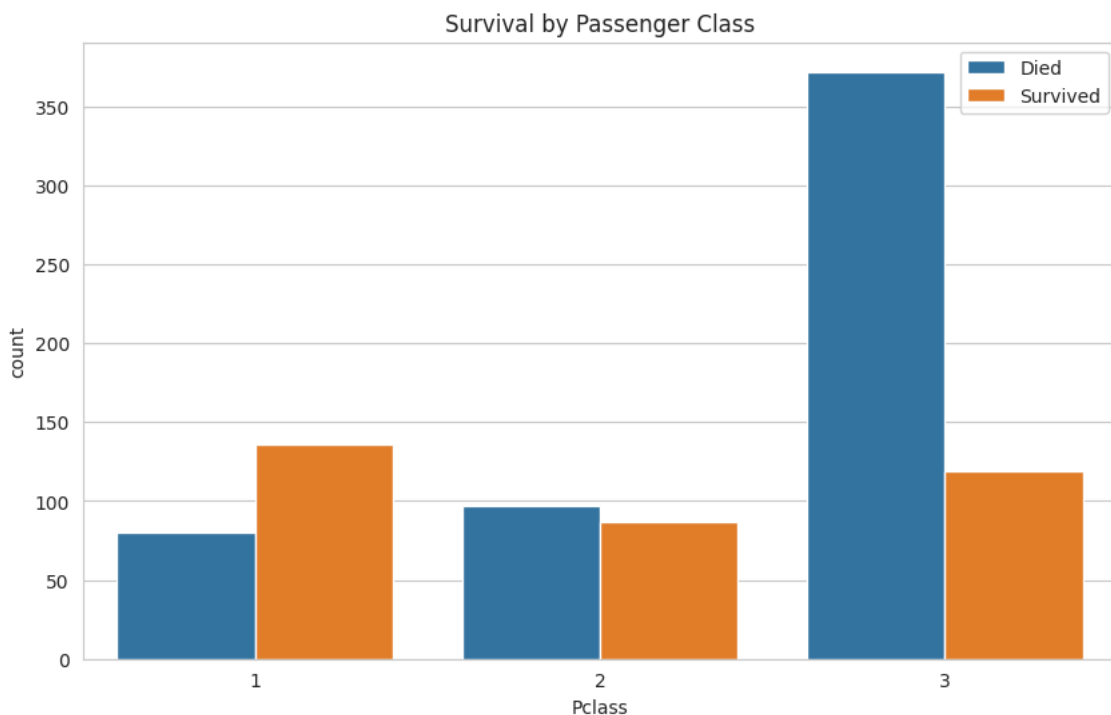
Visualization 1: How many survived?A simple `countplot` is the best way to see the distribution of a categorical variable, like our target `Survived`.

```
sns.set_style('whitegrid') # Sets a nice visual style for our plotsplt.figure(figsize=(8, 6))sns.countplot(x='Survived', data=df)
```

Visualization 2: Does passenger class matter for survival?Now we want to see if there's a relationship between two variables. We can use the `hue` parameter in `countplot` to split the bars by another category.

```
plt.figure(figsize=(10, 6))
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Passenger Class')
plt.legend(['Died', 'Survived'])
plt.show()

print("Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd class passengers")
```



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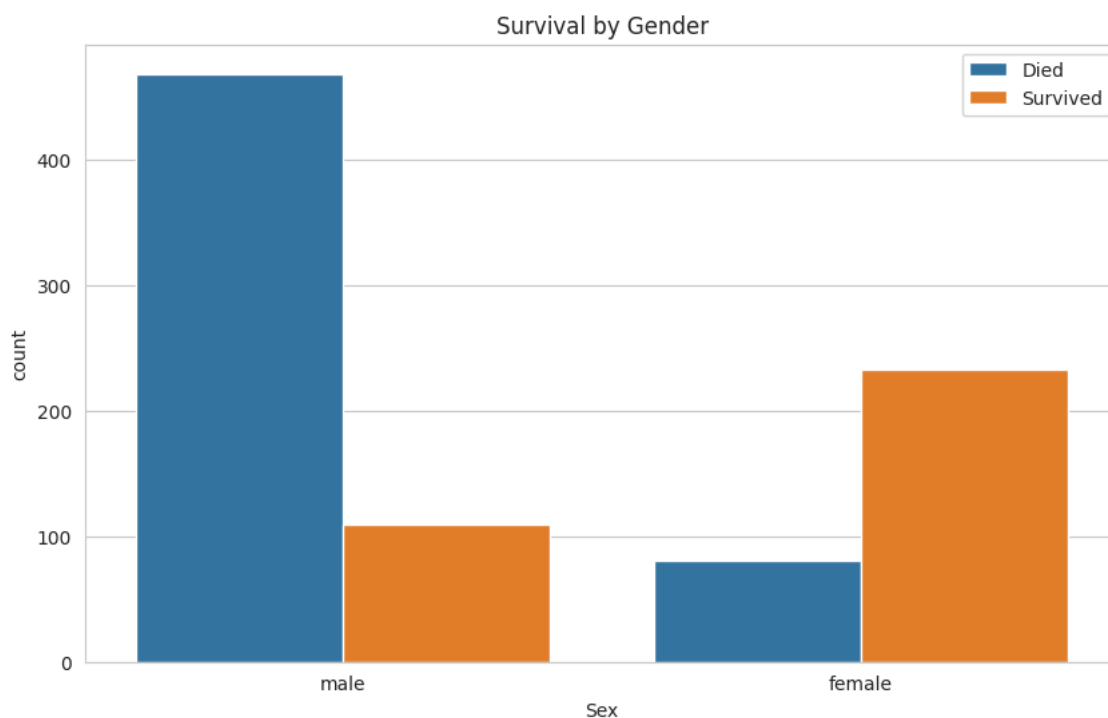
Visualization 3: What about gender?The 'women and children first' mantra is famous. Let's see if the data supports it.

```
plt.figure(figsize=(10, 6))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Gender')
```

```
plt.legend(['Died', 'Survived'])

plt.show()

print("Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is another very strong pattern in the data.")
```



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Visualization 4: How does age play a role? For a continuous variable like **Age**, a histogram is a great way to see its distribution.

A FacetGrid allows us to create multiple plots side-by-side to compare distributions. Here, we create one histogram for passengers who survived and one for those who did not.

Part 4: Student Experimentation **Instructions:** Create your own visualizations to explore other relationships in the data.

Experiment 1: Port of Embarkation. The **Embarked** column tells you where a passenger boarded the ship (C = Cherbourg, Q = Queenstown, S = Southampton). Create a **countplot** to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

--- ENTER YOUR CODE HERE ---

Experiment 2: Fare vs. Survival. **Fare** is a continuous numerical variable. A **boxplot** or **violinplot** is a great way to see the distribution of fares for those who survived vs. those who didn't. Create one of these plots to compare the **Fare** distribution by **Survived**. What does it tell you?

--- ENTER YOUR CODE HERE ---

Close

Knowledge Check **Instructions:** Answer the following questions in this markdown cell.

- What is the primary goal of Exploratory Data Analysis (EDA)?
- Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic? (Describe them in terms of class, gender, and age).
- Why is it important to visualize data instead of just looking at summary statistics? What can a plot show you that a number like 'mean' can't? [ENTER YOUR ANSWERS HERE]

The main goal of EDA is to understand the data, find patterns, identify relationships, and notice any problems before using the data for analysis or machine learning.

Based on the plots, a female passenger in 1st class, especially a young passenger, had one of the best chances of surviving. The charts show a strong relationship between survival and gender, class, and age.

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- Why is it important to visualize data instead of just looking at summary statistics? What can a plot show you that a number like 'mean' or 'count' can't? [ENTER YOUR ANSWERS HERE]

gender and passenger class had a strong relationship with survival. Visualizations make patterns and relationships easier to see. A mean gives you a number, but a plot can show differences between group distributions, and unusual patterns that a single number might hide.

The main goal of EDA is to understand the data, find patterns, identify relationships, and notice any problems before using the data for further analysis or machine learning. Based on the plots, a female passenger in 1st class, especially a younger passenger, had one of the best chances of surviving. The charts show that gender and passenger class had a strong relationship with survival. Visualizations make patterns and relationships easier to see. A mean or count gives you a number, but a plot can show differences between groups, trends, distributions, and unusual patterns that a single number might hide.